Run_55 Micro-TPC 3-D Tracking: Source Pointing, Alignment & Drift Calibration

n_TOF EAR2 X17 — DREAM Micromegas, scint-doubles trigger, ^{3}He target (July 18–19, 2026)

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analysis and slides prepared with Claude (Anthropic AI assistant)

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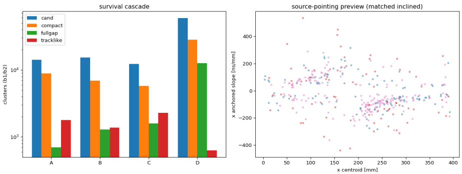
Where we stand — one slide

- From the 24 recovered run_55 subruns (76 214 triggers) a stringent realism gate yields **762 clean X/Y 3-D micro-TPC segments** (A 308, B 153, C 287, D 14).
- The tracks are **real** (bench charge-balance value, tight vs the shuffled null) and they **point back at the ^{3}He source** in all eight planes.
- An **in-situ source-hypothesis calibration** is in hand per plane: transverse alignment (u_0), angle scale (charge sharing), and a fringe-field distortion map. Drift velocity confirmed on chamber C.
- Caveat: the calibration is fit *assuming* the tracks come from the source — it shows self-consistency, not an independent validation. Absolute angle scale / v_{drift} stay partly degenerate; single-track pointing is resolution-limited ($\sim 11^\circ$).

Data local at `~/x17/beam_july/runs/run_55/` (combined hits + decoded waveforms). Scripts `27*` in `mx_july_beam_qa/`; full writeup `TRACKING_RUN55_ANALYSIS.md`.

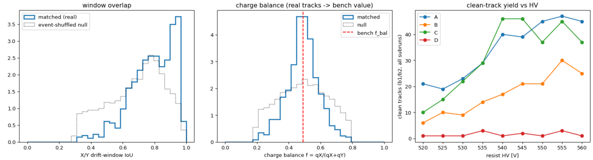
The micro-TPC picture and the realism gate

- Each chamber is a single micro-TPC 30 mm from the target; electrons drift the gap along the radial normal, strips read $(u, v) =$ (transverse, beam-axis).
- A source track $\Rightarrow$ a **monotonic full-gap trail**: strip position vs peak drift-sample is an ordered line spanning the gap.
- Gate (per plane): compact ($\leq$25 mm), full-gap peak-time span ($\geq$6 samp), trail monotonicity $|\rho| \geq 0.8$; then X/Y consistency in drift-time, length and charge balance.
- **Key discriminator:** use the peak-time span (`dur`), NOT the pulse-envelope — high-gain single pulses fake full-gap otherwise.



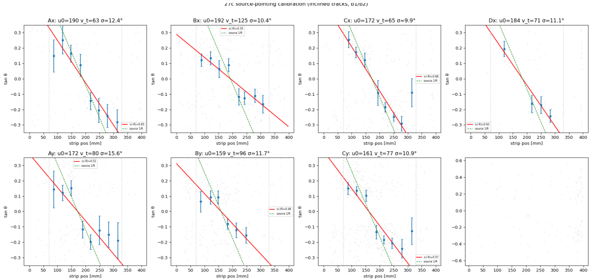
The survival cascade: D's 59k capture-flood candidates collapse to 354 inclined — the gate does the contamination rejection it was built for.

The tracks are real: charge balance vs the shuffled null



- Matched X/Y pairs sit at the June **bench charge-balance** $f = q_X/(q_X+q_Y) \approx 0.48$–$0.50$ and are much **tighter** than the event-shuffled accidental null.
- Random combinatorics cannot reproduce the bench value at that width $\Rightarrow$ the segments are genuine.
- Track-like multiplicity $\approx$1.04 per plane: 90% of two-plane events are a unique 1×1 pairing.
- Clean-track yield rises with resist HV for A/B/C (gain turn-on), flat/low for D — same story as the 25b efficiency proxy, now on verified tracks.

In-situ calibration: alignment, angle scale, distortion

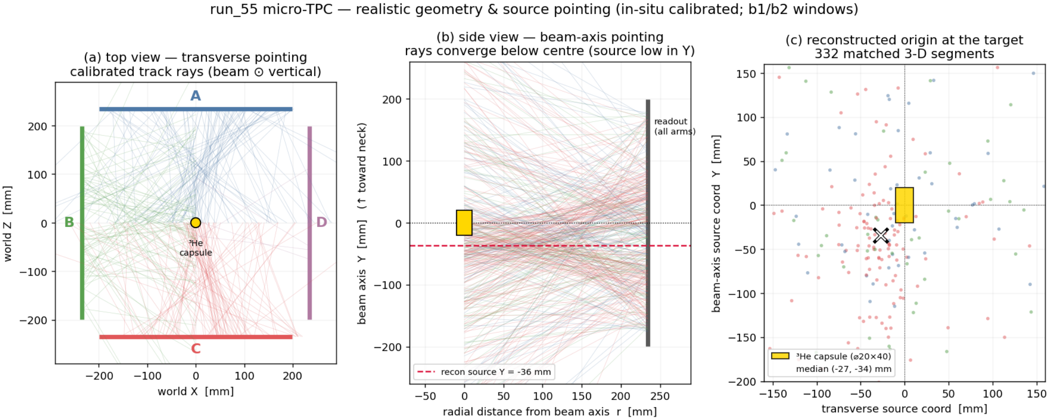


Source model per plane: $\tan\theta = (u - u_0)/R$, $R \approx 234$ mm. Fit $\tan\theta_{\mathrm{meas}}$ vs u in a 70–330 mm fiducial:

- **Alignment** u_0: transverse (x) good to $\sim$**1 cm** on all chambers.
- **Angle scale** $s \cdot (-R) \approx$ **0.62**, consistent A/C/D — the $\sim$40% charge-sharing *compression* of the raw time-fit (same bias the bench regression removes), NOT a v_{drift} error. B is the outlier (0.35; wide-cluster/ringing pathology).
- **Edge distortion** (next slide): residual flat in-fiducial, turns up $\sim$+10° beyond 330 mm in *every* plane ⇒ a real drift fringe field, mapped in `calib/27_align.json`.

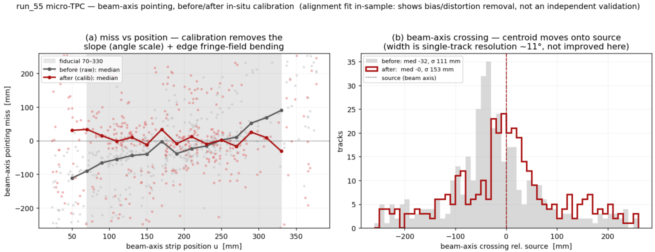
Stored: `27_align.json` (u_0, `scale_sR`, `dist_pos/dtan`, v_{nom}).

Realistic geometry & source pointing



(a) Pinwheel of the 4 chambers (A top / C bottom / B left / D right), beam vertical; calibrated rays fan back onto the ^{3}He capsule. **(b)** Side view: rays converge *below* centre. **(c)** 332 matched segments; reconstructed origin median $(-27, -34)$ **mm** vs the $\varnothing 20 \times 40$ capsule. Convergence is statistical — single-track pointing $\sim 11^\circ$.

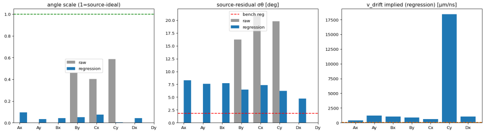
Before / after calibration on the beam axis



- **(a)** Raw pointing miss slopes $-110 \to +90$ mm across the strip (angle-scale undershoot + edge bending); calibrated is **flat on zero** through the fiducial.
- **(b)** Beam-axis crossing centroid moves $-32 \to 0$ **mm** — onto the source.
- Honest read: calibration removes *bias* and *position-dependent distortion*. It does *not* shrink the width (σ 111→153 mm): single-track pointing is set by the $\sim 11^\circ$ resolution (extended source $\sim 5^\circ$, near-normal short lever, coherent CM noise).
- The fit is in-sample, so this is a self-consistency / bias-&-distortion demonstration, not an independent validation.

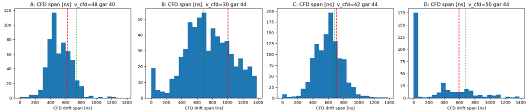
Two physics results fall out

1. The source is ∼3 cm low in beam-Y.



Chambers independently triangulate the same offset (A −21, B −29, C −48 mm). At EAR2 the beam is vertical and neutrons enter the capsule from below, so captures pile up at the **bottom** of the $D20 \times L40$ capsule — this is real source position, NOT a y-misalignment to correct.

2. Drift velocity, telescope-free.



Full-gap drift-time span = gap/v, angle-independent. From CFD waveform reconstruction (2701 tracks): **C sits on garfield** (42 vs 44 μm/ns); A/D read ∼10% high, B low (untrusted). $\Rightarrow v_{\mathrm{drift}} \approx 44$ μm/ns is consistent for the trusted chamber; the ∼0.6 angle scale is charge sharing, not v.

What does NOT work (yet), and why

- **The frozen bench angle-regressor does not transfer.** Restandardized onto run_55 features, the bench $|\tan|$ model (holdout σ 1.7–1.8° on cosmics) collapses — source-pointing slope $\to 0$. Cause: it was trained on cosmics spanning a broad angle range; run_55 source tracks are **near-normal incidence** (small, narrow tan), so restandardization mis-maps the feature scale.
- **Sub-sample timing does not help.** CFD reconstruction of all 2701 inclined tracks gives the *same* ~0.6 compression and the *same* ~11° resolution as the 60 ns max-sample fit $\Rightarrow$ the compression is genuine charge sharing and the resolution is S/N-limited (coherent CM noise on low-amplitude trail strips), not timing-quantisation-limited.
- **Everything is b1/b2 windows only** (8–12, 16–28 ms); the 2–6 ms physics target is unsampled (the DAQ FIFO comb).

Are we calibrated?

First-order yes — and here is exactly what that means:

- ✓ **Alignment**: transverse u_0 to $\sim$1 cm, all chambers.
- ✓ **Angle scale**: 0.62 charge-sharing compression measured and consistent A/C/D.
- ✓ **Distortion**: edge fringe-field map per plane; fiducial is clean.
- ✓ **Drift velocity**: garfield confirmed on chamber C.
- ✓ **Pointing**: unbiased; centroid nails the source including the -3 cm beam-Y physics.

But not closed:

- ▷ The calibration is **in-sample** (source-assumed) — self-consistent, not independently validated.
- ▷ Absolute angle scale $\leftrightarrow v_{\mathrm{drift}}$ remain **partly degenerate** (frozen transfer failed).
- ▷ Single-track resolution stuck at $\sim 11^\circ$ ($\sim$8–15 cm at the target).
- ▷ B chamber not trusted for scale or v.

Follow-ups (in priority)

1. **Re-train the angle regressor in situ** on run_55 tracks with source-truth labels ($\tan\theta = (u - u_0)/R$), holding out for an honest residual σ_θ — this both removes the charge-sharing compression and cleanly separates v_{drift}.
2. **Improve S/N before precision.** The $\sim$11° resolution is coherent-CM-noise-limited on low-amplitude trail strips; reprocess with CM correction / ZS ($=1$, 3.5σ from the ZS study) to sharpen the trail and v_{drift}.
3. **Use the low-Y source as the truth anchor** in inter-chamber linking / e^+e^- vertexing (ntof_tracking PLAN_04); apply the distortion map or fiducialize to 70–330 mm.
4. **Diagnose B** wide-cluster/ringing at the waveform level (also open from the HV scan).

Figures: `figures/27_tracks/` (this deck adds `14_pointing_geometry`, `15_beamaxis_convergence` via `27g_pointing_geometry.py`). Calibration: `calib/27_align.json`, tracks `calib/27_tracks.npz`.